

**FROM MEMBRANE PERTURBATION TO BIOFILM ARCHITECTURE
- IMPACT OF PSMA3 ON *STAPHYLOCOCCUS EPIDERMIDIS* -**

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Keywords: Biofilm, staphylococci, membrane, cell wall, and Phenol-Soluble Modulins

ABSTRACT

Polymicrobial biofilms formed by *Staphylococcus aureus* and *Staphylococcus epidermidis* are a major clinical challenge due to their antibiotic resistance¹. Within these communities, bacteria secrete virulence factors, such as phenol-soluble modulins (PSMs), which play key roles in biofilm regulation, and eventually in niche competition². Here, we investigate the impact of PSM α 3 from *S. aureus* on the competing bacterium *S. epidermidis*, using a multiscale *in vitro* and *in cellulo* approach, combining calcein leakage assays, viability tests, atomic force microscopy, and confocal imaging.

At the cellular scale, our results reveal that PSM α 3 leads to a 3 fold decrease in bacteria elasticity without a major loss of viability, suggesting sublethal effects. Besides, on simplified models (in absence of cell wall components), PSM α 3 induces concentration- and lipid-dependent membrane perturbations. Importantly, at the community level, microbiological analyses suggest that PSM α 3 could promote dispersion of bacterial aggregates rather than direct bactericidal effect. This hypothesis is further supported by biofilms imaging, indicating that PSM α 3 might alter biofilm integrity, through modulation of its thickness and density. Overall, by integrating biophysical tools across scales, from molecular interactions to bacterial ensemble dynamics, we highlight subtle effects of PSM α 3 from *S. aureus* on *S. epidermidis*, that may influence microbial competition.

REFERENCES

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